

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented PLC programming, robotics integration, and vision system configurations for automated production lines, ensuring adherence to **Machinery Directive** and **CE compliance**.
- Troubleshoot and resolved complex automation issues on live production equipment, including detailed interrogation of **PLC code** to identify root causes and minimize downtime.
- Collaborated with cross-functional teams to develop and implement effective repairs and system improvements, contributing to enhanced machine productivity and reliability.
- Managed the design, installation, and modification of automation equipment, including **PLCs**, servo drives, and SCADA systems, providing technical recommendations for hardware and architecture selection.
- Developed a critical spares list for site controls and automation hardware, negotiating and maintaining associated support contracts to ensure consistent availability and minimize risk.
- Coached junior control engineers and area technicians on automation troubleshooting techniques and continuous improvement methodologies, enhancing overall team technical capability.
- Containerized key automation services using **Docker** and deployed them to a managed **Kubernetes** cluster on **AWS**, ensuring predictable rollouts and resource utilization.

SKILLS

Automation & Controls – PLC Programming, Robotics, Vision Systems, SCADA, Servo Drives, Motion Control, Allen-Bradley (Rockwell), Siemens S7, Telemecanique, EtherNet IP, DeviceNet, ControlNet, Modbus, Profibus, GAMP, URS, FDS, FAT, SAT, Machinery Directive, CE Compliance, EN 60204-1, Electrical Control Panels

Software Development – Java, Spring Boot, REST APIs, Python, SQL, Hibernate, JPA, JUnit, Docker, Git, Jenkins

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Kubernetes, Docker, CI/CD, Jenkins

Data & Analytics – Power BI, Tableau, DAX, Power Query, pandas, NumPy, R, RStudio, Statistical Analysis

Operations & Support – Production Support, Incident Response, SLA Management, Change Management, Root Cause Analysis (RCA), Splunk, Dynatrace, Linux Shell Scripting

PROJECTS

Automated Pharmaceutical Packaging Line Control System

- Designed and programmed the control logic for a multi-stage pharmaceutical packaging line using **Siemens S7 PLCs** and integrated **Cognex vision systems** for quality inspection.
- Developed fault-finding routines and diagnostic dashboards in **Python** to assist operators in quickly identifying and resolving issues, reducing average troubleshooting time by approximately **18%**.
- Managed the installation and commissioning of **servo drive** and motion control systems for robotic pick-and-place units, ensuring precise operation within tight tolerances.

SCADA System Enhancement for Manufacturing Data Collection

- Enhanced the site's **SCADA system** to collect real-time operational data from 50+ production machines, feeding into a central **PostgreSQL** database for analysis.
- Developed automated **Python scripts** to validate and pre-process incoming SCADA data, identifying and flagging anomalous readings with a **92%** accuracy rate before database ingestion.
- Built interactive **Power BI dashboards** to visualize key performance indicators derived from the SCADA data, providing operations teams with actionable insights into machine uptime and throughput.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)